

DR-2 for the lattice class – an UNCONDITIONAL decoupling-free closure

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The configurations DR-2 must handle in TECT are BCC/FCC reciprocal-lattice momentum shells – lattice points on a sphere – not adversarial sets. v1.0 reduced DR-2 for this class to the named classical hypothesis [DIV-CIRC] (lattice points on a circle number $O_\epsilon(R^\epsilon)$) and graded it T6 CONDITIONAL. This v1.1 PROVES [DIV-CIRC] in full via a shift-removing substitution, upgrading the result to T7 UNCONDITIONAL (modulo two textbook number-theory facts). The route is self-contained: no Bourgain–Demeter machinery, no R1/R2 residuals, no conjecture.

2. The lattice reduction and the proof of [DIV-CIRC]

Let $\Lambda = \mathbb{Z}^3$ (the argument is lattice-independent) and $Q = \Lambda \cap \{x : |x|^2 = R\}$, $N = |Q|$. By Lemma A (R-025), $E_+(Q) \leq (1 + T'(Q))N^2$ with $T'(Q) = \max_{m \neq 0, r(m) > 0} \#(Q \cap C_m)$ and $C_m = \{x : |x|^2 = R, x \cdot m = \frac{1}{2}|m|^2\}$. We bound $\#(\mathbb{Z}^3 \cap C_m)$.

Lemma [DIV-CIRC]. For $m \in \mathbb{Z}^3$, $0 < |m|^2 \leq 4R$,

$$\#(\mathbb{Z}^3 \cap C_m) \leq 6d(4R - |m|^2) \ll_\epsilon R^\epsilon,$$

where $d(\cdot)$ is the divisor function, uniformly in m .

Proof. (i) Shift removal. The circle C_m has centre $m/2$ (the midpoint of every pair $a+b=m$). For $x \in \mathbb{Z}^3 \cap C_m$ set $y := 2x - m$. Then $y \in \mathbb{Z}^3$ and

$$y \cdot m = 2x \cdot m - |m|^2 = 2 \cdot \frac{1}{2}|m|^2 - |m|^2 = 0,$$

so $y \in \Lambda_m := \mathbb{Z}^3 \cap m^\perp$, a rank-2 sublattice. Moreover

$$|y|^2 = 4|x|^2 - 4x \cdot m + |m|^2 = 4R - 2|m|^2 + |m|^2 = 4R - |m|^2 =: R'.$$

The map $x \mapsto y$ is injective ($x = (y + m)/2$). Hence

$$\#(\mathbb{Z}^3 \cap C_m) \leq \#\{y \in \Lambda_m : |y|^2 = R'\} =: r_Q(R'),$$

where $Q(y) = |y|^2$ is the positive-definite INTEGRAL binary quadratic form induced on Λ_m (integer Gram matrix, $\Lambda_m \subset \mathbb{Z}^3$). This is a HOMOGENEOUS (origin-centred) representation count – the centre $m/2$ has been removed exactly.

(ii) Class-number bound. For a primitive positive-definite integral binary quadratic form of discriminant Δ , Dirichlet's class-number formula gives

$$\sum_{[f] \text{ disc } \Delta} r_f(n) = w(\Delta) \sum_{d|n} \chi_\Delta(d) \leq 6d(n),$$

where $w(\Delta) \leq 6$ is the number of units and $\chi_\Delta \leq 1$. A single class's count is at most the sum over all classes, so $r_Q(R') \leq 6 d(R')$ (if $Q = g Q'$ is imprimitive of content g , $r_Q(R') = r_{Q'}(R'/g) \leq 6 d(R'/g) \leq 6 d(R')$). This bound is UNIFORM in m (independent of Δ , hence of the plane).

(iii) **Divisor bound.** $d(R') \ll_\epsilon R'^\epsilon \leq (4R)^\epsilon \ll_\epsilon R^\epsilon$. $\square$

Both inputs – the class-number formula and $d(n) \ll_\epsilon n^\epsilon$ – are textbook; [DIV-CIRC] is therefore a theorem, not a hypothesis.

3. Theorem (lattice DR-2, unconditional)

Theorem. For $Q = \Lambda \cap \{|x|^2 = R\}$ (any fixed lattice Λ),

$$E_+(Q) \leq \left(1 + 6 \max_{m \neq 0} d(4R - |m|^2)\right) N^2 \leq (1 + C_\epsilon R^\epsilon) N^2.$$

For Gauss-typical shells $R \asymp N^{2+o(1)}$, $E_+(Q) \lesssim_\epsilon N^{2+\epsilon}$ – DR-2 holds for the lattice class, UNCONDITIONALLY and with NO decoupling.

Proof. Lemma A gives $E_+(Q) \leq (1 + T'(Q)) N^2$, and [DIV-CIRC] gives $T'(Q) = \max_{m \neq 0} \#(\mathbb{Z}^3 \cap C_m) \leq 6 \max_{m \neq 0} d(4R - |m|^2) \ll_\epsilon R^\epsilon$. For $R \asymp N^2$, $R^\epsilon \asymp N^{2\epsilon}$. $\square$

The two statements are kept separate: the boxed bound holds for ALL lattice shells; the $N^{2+\epsilon}$ form requires the absorption $R \leq N^{O(1)}$, which holds for Gauss-typical (dominant) shells but not for very sparse shells (where the absolute energy is small in any case).

4. Numerical verification (exact arithmetic)

Proof chain (dr2_divcirc_proof.py, 4/4, exact integers). On the richest PROPER sum-circle of each shell $R = 101, 314, 561, 749$: the substitution $y = 2x - m$ lands in Λ_m with $|y|^2 = 4R - |m|^2$ EXACTLY and is injective; $\#(\mathbb{Z}^3 \cap C_m) = r_Q(R')$ (the homogeneous count); and $r_Q(R') \leq 6 d(R')$ with margin ($\max T'/6d = 0.25$). The degenerate $m = 0$ (whole-sphere antipodal) case is excluded, exactly as the proof splits it off.

Growth (dr2_lattice_divisor.py, 5/5). On $\mathbb{Z}^3$ shells $R = 101, \dots, 9974$ ($N = 168, \dots, 2040$): the Lemma A assert holds; T'/N falls $0.107 \rightarrow 0.024$; the log-log slope of T' vs R is 0.177 (the $R^{o(1)}$ signature); $E_+/N^2 \leq 5.3$; $\mathbb{Z}^3 = \text{FCC}$ at $R = 1826$ (lattice-independent).

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

Four concrete objections, found before submission:

- (α) “Is the class-number bound really uniform over the varying planes?” YES, and this is the crux of why v1.1 is T7. The bound $r_Q(R') \leq 6 d(R')$ comes from “single class $\leq$ sum over all classes of the same discriminant $= w \sum \chi(d) \leq 6 d(R')$ ”; the right-hand side has NO discriminant dependence, so it is uniform in m . Verified: $T'/6d \leq 0.25$ on four shells.
- (β) “The substitution might not be injective or might leave Λ_m .” Checked exactly: $y \cdot m = 0$ (so $y \in \Lambda_m$) and $|y|^2 = 4R - |m|^2$ hold for every x on the richest circle, and $x = (y + m)/2$ recovers x (injective). The parity coset $y \equiv m \pmod{2}$ is dropped (an upper bound).

- (γ) “Does this close DR2-SHARE / discharge H-ADM-COH?” NO, and this is the honest scope limit. The note closes the ADDITIVE-ENERGY DR-2 for the lattice class. The STEP-5B carrier richness $\chi(P)$ and T' are both rich-circle quantities, closely related, but their exact equivalence is NOT established here; the discharge of H-ADM-COH is a separate OPERATOR integration decision. NO gate/hypothesis flip is made.
- (δ) “Arbitrary-Q DR-2?” STILL OPEN. This route uses the arithmetic of Λ essentially (the rich-latitude example with $T' = N$ shows non-lattice Q is not reached). The decoupling route (T4+) remains the general-case path.

Quantitative sanity check (mandatory). Exact integer arithmetic. Substitution: $|y|^2 = 4R - |m|^2$ verified per point. Class-number: $r_Q(R') \leq 6d(R')$, $T'/6d \leq 0.25$. Magnitude: $E_+/N^2 \leq 5.3$ ($\ll N$). Growth: $T'/N \rightarrow 0$, log-log slope $0.177 < 1$. Limit: $m = 0$ excluded (split off as $r(0)^2 \leq N^2$); $\mathbb{Z}^3 = \text{FCC}$ at $R = 1826$. Direction: upper bound, all terms positive.

6. Result footer

Result ID:	R-026 / DR-2 for the lattice class (Route A, T7)
Precise statement:	For $Q = \Lambda \cap \{ x ^2 = R\}$ (Λ a fixed lattice), $E_+(Q) \leq (1 + 6 \max_{m \neq 0} d(4R - m ^2)) N^2$ $\leq (1 + C_{\text{eps}} R^{\text{eps}}) N^2$, by Lemma A (R-025) and [DIV-CIRC]. The latter is PROVED: $x \mapsto y = 2x - m$ sends $\mathbb{Z}^3 \cap C_m$ injectively into $\{y \in \Lambda_m : y ^2 = 4R - m ^2\}$, a homogeneous rank-2 representation count $\leq 6 d(4R - m ^2)$ (Dirichlet class-number formula, uniform) $\ll_{\text{eps}} R^{\text{eps}}$. For $R \sim N^2$, $E_+(Q) \leq_{\text{eps}} N^{2+\text{eps}}$. DECOUPLING-FREE.
Scope:	TECT DR-2 carrier (BCC/FCC momentum-shell lattice points); additive-energy form of DR-2.
Dependencies:	Lemma A (R-025, unconditional) + two TEXTBOOK facts: Dirichlet class-number formula and $d(n) \ll_{\text{eps}} n^{\text{eps}}$. NO decoupling, NO conjecture.
Evidence grade:	T7 PROVED UNCONDITIONAL (modulo textbook number theory; operator pre-authorised T7 on full [DIV-CIRC] expansion). v1.0 T6 CONDITIONAL superseded.
Reproduction command:	python codes/vacuum/dr2_divcirc_proof.py python codes/vacuum/dr2_lattice_divisor.py
Expected output:	divcirc_proof 4/4 (substitution exact; $T' \leq r_Q(R')$ $\leq 6 d(R')$; $T'/6d \leq 0.25$); lattice_divisor 5/5 (Lemma A holds; $T'/N \rightarrow 0.107 > 0.024$; $E_+/N^2 \leq 5.3$).
Falsification gate:	A lattice shell with $E_+ > (1+T')N^2$ (refutes Lemma A), or $\#(\mathbb{Z}^3 \cap C_m) > 6 d(4R - m ^2)$ (refutes [DIV-CIRC]); neither observed.
Tier before / after:	v1.0 T6 CONDITIONAL $\rightarrow$ v1.1 T7 UNCONDITIONAL for the LATTICE class. General/arbitrary-Q DR-2 unchanged (T4+). NO flip: DR2-SHARE OPEN, B5 T5, B1 T6, H-ADM-COH retained.
No-overclaim statement:	Closes the additive-energy DR-2 for the LATTICE class only (T7). Does NOT close arbitrary-Q DR-2 and does NOT by itself discharge H-ADM-COH; the T' -to- $\chi(P)$ equivalence and any gate/hypothesis flip are operator decisions.
Next required action:	Operator integration: establish $\chi(P) \sim T'$ (the STEP-5B carrier-richness $\leftrightarrow$ sum-circle-richness link), then decide whether the lattice-class T7 closure discharges H-ADM-COH in B1.